Review notes · for you

Train your own model: unit check

AI · Level 3 · Unit check

Materials
Paper and a pen. No device needed.

Learning goals

- Data first
- How a classifier decides
- Test it honestly
- Fair for everyone

Activity: Team quiz

1. Make teams of three or four, or quiz a family member
2. Read each question aloud; agree on one answer
3. Reveal the answer and explain why it is right
4. Write the hardest question in your notebook to revise

Discussion

- Which idea from this unit surprised you most?
- What would you check first before trusting an AI tool's score?
- What model would you like to build, and whose data would it need?

Check yourself

- One question I will ask about the next AI tool I use

Try it this week
Pick one AI tool you use (voice typing, a translator, a photo app) and ask the three questions from this unit: what data did it learn from, how was it tested, and who might it fail?

Quiz sheet

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Name _____ Class _____ Date _____

1 **You want a model that tells healthy from sick cassava leaves. Which dataset is best?**

- ◯ 100 healthy photos and 5 sick ones
- ◯ Any leaf photos from the internet, unlabelled
- ◯ 60 healthy and 60 sick, all from one plant at noon
- ◯ 30 healthy and 30 sick, from different fields, times and phones

2 **A model treats every label in its training data as correct, even the wrong ones.**

◯ True ◯ False

3 **Match each word to its meaning.**

Feature	• • • •	Examples kept apart to measure the model honestly
Label	• • • •	A number you can measure about every example
Test set	• • • •	More mistakes for one group than for others
Bias	• • • •	The right answer given to an example

4 **A new mango's nearest neighbour is ripe, but its next four neighbours are all unripe. What does the classifier say with k = 1, and with k = 5?**

- ◯ Ripe both times
- ◯ Not ripe both times
- ◯ It can't decide with k = 5
- ◯ Ripe with k = 1, not ripe with k = 5

5 **Put the steps of building and checking a model in order.**

- ☐ Collect examples fairly, with permission
- ☐ Label them and check the labels
- ☐ Test on the test set, group by group
- ☐ Plan the data: what, labels, how many, how varied
- ☐ Train on most examples, keeping a test set apart

6 **Which of these are signs of overfitting, or make it likely? Choose all that apply.**

- ☐ Similar scores on training and test examples
- ☐ A test on examples it never saw
- ☐ Lots of freedom, trained on very few examples
- ☐ A curve that passes through every training point but swings wildly between them
- ☐ 100% on training examples, 60% on new ones

7 **Your club's model reads handwritten exam numbers. It is 92% right overall, but only 70% right for the youngest class. What should you do?**

- ◯ Collect samples from the youngest class with permission, retrain, test each class
- ◯ Publish 92%: it is the true overall score
- ◯ Ask the youngest class to write like the older students
- ◯ Use it only for older classes and say nothing

8 **A model with a 95% overall score can still get most examples from one small group wrong.**

◯ True ◯ False

9 **Complete the sentence.**

Choose _____ that separate the classes, keep a _____ set apart, and check each _____ separately.

Word bank: colours · features · training · group · test

10 **A model looks for sick cows in 20 test photos. Ten cows are really sick. It calls 9 cows sick: 7 of them are sick and 2 are healthy. How many sick cows did it miss (false negatives)?**

◯ 7 ◯ 2 ◯ 11 ◯ 3

Check your answers

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1 **30 healthy and 30 sick, from different fields, times and phones**

Thirty of each from different fields, times and phones is balanced, labelled and varied.

2 **True**

A model can't tell a wrong label from a right one: it learns from all of them.

3 **Feature — A number you can measure about every example**
Label — The right answer given to an example
Test set — Examples kept apart to measure the model honestly
Bias — More mistakes for one group than for others

Features describe examples, labels give the answers, the test set measures the model, and bias is what testing group by group can reveal.

4 **Ripe with k = 1, not ripe with k = 5**

With k = 1 only the closest mango votes, and it is ripe.

5 **Top to bottom: 2, 3, 5, 1, 4**

Planning comes before collecting, and labels before training.

6 **100% on training examples, 60% on new ones; A curve that passes through every training point but swings wildly between them; Lots of freedom, trained on very few examples**

A big gap between training and test scores, a curve that memorises every point, and lots of freedom with few examples all point to overfitting.

7 **Collect samples from the youngest class with permission, retrain, test each class**

The 70% shows a data gap: the youngest class's writing was missing or rare.

8 **True**

If a group makes up only a few of the test examples, the model can fail most of them and the overall score barely moves.

9 **Choose features that separate the classes, keep a test set apart, and check each group separately.**

Good features, a test set the model never saw, and group-by-group checks: the three habits of an honest model maker.

10 **3**

Ten cows are sick and it caught seven, so it missed three: three false negatives.